Why does depth, not shape, set the pressure?

? Pressure is called a scalar, yet it pushes on every surface. Where does the direction come from?

1 ## A point has no orientation

Cut a small wedge of fluid with a slanted face dA at angle θ. The three faces have areas dA, dA sinθ and dA cosθ. Resolving along the slant and across it gives the same pressure on all three.

$$p_x = p_y = p_\theta \tag{3.1}$$

→ shrink the wedge: the weight falls as dV, the forces only as dA, so the weight drops out

2 ## The vertical gradient is a slab weight

For a slab of area A and thickness dy, the net pressure force must carry its weight. With y measured upward:

$$\frac{dp}{dy} = -\rho g$$

Density may vary with height; for a liquid it does not, and the equation integrates at once.

3 ## Integrate once and you own hydrostatics

Take h downward from the free surface, where the pressure is p_0. Constant density gives the everyday formula, and a layered fluid is integrated layer by layer.

$$p = p_0 + \rho g h \tag{3.2}$$

→ a two-layer tank has a kink in p(h) at the interface, never a jump

4 ## Absolute, gauge, and the barometer

Gauge pressure is p − p_0, the part a manometer can see. A barometer is the same equation solved for the height that balances the atmosphere: 0.76 m of mercury, or 10.3 m of water.

$$p_0 = \rho_{Hg}\, g(0.76) = 1.013 \times 10^5 \text{ Pa}$$

5 ## Shape cannot matter, but weight must go somewhere

Every vessel with the same depth has the same pressure at the base, hence the same force pA on the base, flared or narrow. The walls supply the difference, which is why a flared vessel can weigh much more than the force on its base.

→ hydrostatic paradox: pressure is local, weight is global

KEY RESULT

the three you use

$$p = p_0 + \rho g h$$

$$dp/dy = -\rho g$$

F = pA on any face, along its normal

TRAP

two silent errors

Mixing gauge and absolute inside a Δp balance.

Reading "more water" as "more pressure on the base".

NUMBERS

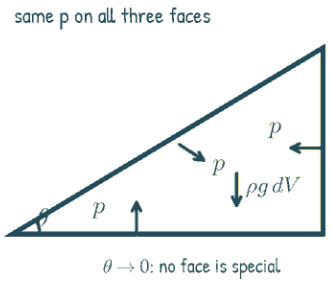
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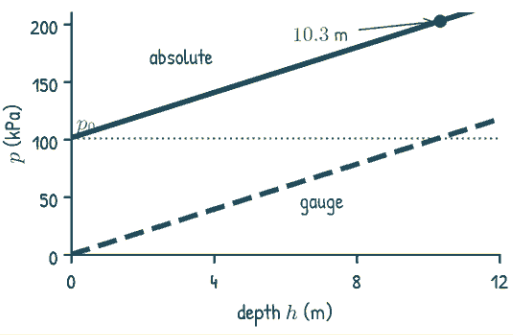
p_0 = 1.013×10⁵ Pa

0.76 m Hg = 10.3 m water

ρHg/ρw = 13.6

Pressure is a scalar that grows with depth; only the area vector gives it a direction





p is one number at a point

$dp/dy = -\rho g$

$p = p_0 + \rho g h$

$F = p\, A\, \hat{n}$

gauge = absolute − p_0

label every pressure

KEEP / pressure is one number per point; direction belongs to the area you choose

How do you walk a U-tube without writing force balances?

? Multi-fluid columns look like bookkeeping. There is one rule, and it survives acceleration.

1 **The step rule**

Start at a point of known pressure, usually an open end at $p_\circ$. Move through the connected fluid: each descent Δh adds $\rho g \Delta h$, each rise subtracts it, and a horizontal run costs nothing.

$$p_{end} = p_0 + \sum_{down} \rho g\, \Delta h - \sum_{up} \rho g\, \Delta h$$

2 **A trapped gas, walked**

Left limb open, 10 cm of water above the mercury; the right mercury stands 18 cm higher over trapped gas. Down the water, up the mercury:

$$p_{gas} = p_0 + \rho_w g(0.10) - \rho_{Hg} g(0.18) = 7.83 \times 10^4 \text{ Pa}$$

→ below atmospheric, which is why the right column stands higher

3 **The barometer is the same walk**

Close one end, fill with mercury, invert into a dish. The column stops when $\rho_{Hg}\, g\, h$ balances $p_\circ$. Water needs 10.3 m, so a water barometer is a building, not an instrument.

$$h = \frac{p_0}{\rho g}$$

4 **Pascal, and what a lift costs**

At the same height in a connected fluid at rest the pressure is equal. A small piston therefore holds a big load with a small force, but the strokes are in the same ratio and the works agree:

$$F_1/a = F_2/A, \qquad F_1 d_1 = F_2 d_2$$

→ a 392 N push over 5 m lifts 19600 N by 0.10 m

5 **A worked reading**

Q3: 10 cm of oil (800 kg/m³) floats on water in one limb of an open U-tube. Balance the two columns at the interface: the water level in that limb sits lower.

$$\rho_o h_o = \rho_w h_w \;\Rightarrow\; h_w = 0.8(10) = 8 \text{ cm}$$

→ Q4: a/A = 1/50 holding 500 kg needs 98 N and a 1 m input stroke for a 2 cm rise

GATE

when the walk works

One connected fluid at rest, or rigidly accelerated (use g_eff).

Same height, same pressure only inside the same continuous fluid.

TRAP

three traps

Adding $p_\circ$ twice when the two ends are both open.

Forgetting the water above the mercury, or its density.

Reading a manometer difference as an absolute pressure.

KEY RESULT

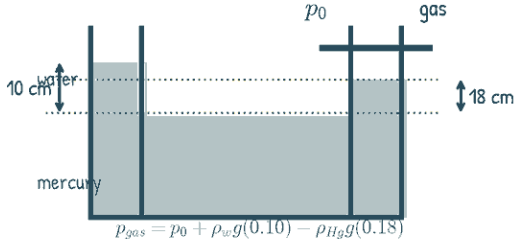
lift audit

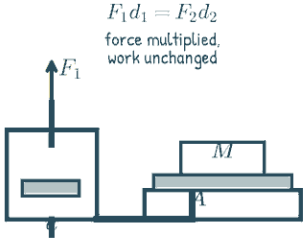
$$F_2 = F_1 A/a$$

$$d_1/d_2 = A/a$$

work in = work out, minus friction

Walk any manometer from an open end: down adds, up subtracts, horizontal costs nothing





KEEP / walk from an open end: down adds, up subtracts, horizontal is free

Where does the push on a gate act?

? A dam knows the total force and the point where a single resultant would act. Both come from the same integral.

1 ## Integrate the linear pressure

For a plane face, p = ρg × (vertical depth), which is linear across the face. The integral of a linear function is its value at the centroid:

$$F = \int p\,dA = \rho g\, h_c A = p_c A \tag{3.3}$$

→ h_c is the vertical depth of the area centroid, not the depth of the plate centre in general

2 ## The resultant sits lower

Taking moments about the centroid gives the centre of pressure. For a rectangle whose top edge is at the free surface, h_c = H/2 and I_c = bH³/12, so h_cp = 2H/3. The extra H/6 is the whole story of a gate hinge.

$$h_{cp} = h_c + \frac{I_c}{A\,h_c} \quad \Rightarrow \quad h_{cp} = \frac{2H}{3}$$

3 ## Inclined faces

Depth is still vertical: h = y sin α from the surface, and F = ρg h_c A. The centre of pressure moves closer to the centroid as the plate approaches horizontal, because the depth variation across it shrinks.

→ torque about a top hinge = F × (h_cp − h_hinge)

4 ## Shape independence, and where the weight went

Three vessels with the same base and depth have equal base forces, though their weights differ. The difference is carried by the walls: a flared vessel pushes down on its base less than it weighs.

5 ## A worked gate

A 2 m high, 3 m wide gate hangs with its top edge at the water surface. The centroid is 1 m down, and the resultant acts 2H/3 below the surface.

$$F = 5.88 \times 10^4\ \text{N}, \qquad \tau_{top} = F(4/3) = 7.84 \times 10^4\ \text{N}\cdot\text{m}$$

→ the hinge feels the centre of pressure, not the centroid: the lever arm is 4/3 m, not 1 m

KEY RESULT

plane faces

$$F = p_c A$$

$$h_{cp} = h_c + \frac{I_c}{A h_c}$$

rectangle from the surface: 2H/3

TRAP

hinge mistakes

Putting the resultant at the centroid.

Using the slant length instead of vertical depth.

Forgetting that a free surface at the top means zero pressure there.

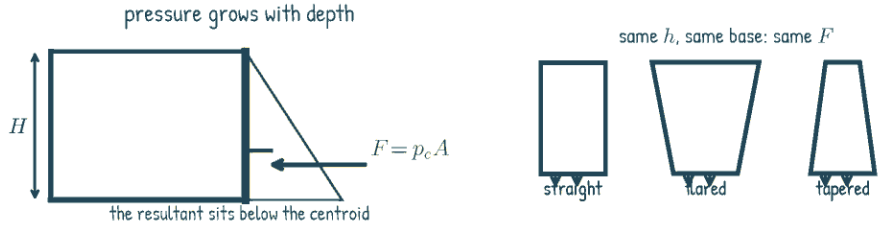
CHECK

sanity

If the plate is horizontal, every point is at one depth: cp = centroid.

If it is deep, cp tends to the centroid because Δp across it is small.

Force on a plane face is the centroid pressure times the area, acting below the centroid



KEEP / force at the centroid of area, line of action below it

How much force does a curved face feel?

? A quadrant gate looks like a calculus problem. Projection turns it into two statics lookups.

1 ## Horizontal: project onto a vertical plane

Every horizontal strip of the curved face has the same depth as the corresponding strip of its vertical projection. So the horizontal resultant equals the force on the projection, applied at the projection's centre of pressure.

$$F_H = p_c A_{proj} \tag{3.5}$$

2 ## Vertical: weigh the column above

The vertical component is the weight of the liquid that would fill the space between the face and the free surface: real column where liquid sits above the face, imagined column where it does not. The sign follows the wetted side.

$$F_V = \rho g V_{column}$$

→ draw the wetted side first; the sign of F_V is decided by that drawing

3 ## Resultant and line of action

Combine the two components. On a circular arc every elemental pressure force is radial, so the resultant must pass through the centre of curvature; use that instead of re-deriving moments.

$$F = \sqrt{F_H^2 + F_V^2}, \qquad \tan\beta = \frac{F_V}{F_H}$$

4 ## The quarter-cylinder gate, numerically

Radius 2 m, width 3 m, water surface at the top of the gate. The projection is 2 × 3 m with its centroid 1 m deep; the column area is the square minus the quarter disc.

$$F_H = 5.88 \times 10^4 \text{ N}, \quad F_V = 2.52 \times 10^4 \text{ N}, \quad F = 6.4 \times 10^4 \text{ N}$$

→ direction 23° below horizontal, line of action through the arc centre

5 ## A worked projection

A hemispherical bump of radius 0.5 m sits in a wall with its centre 2 m below the surface. The horizontal force needs only the projected circle.

$$F_H = p_c A_{proj} = (1000)(9.8)(2)\pi(0.5)^2 = 1.54 \times 10^4 \text{ N}$$

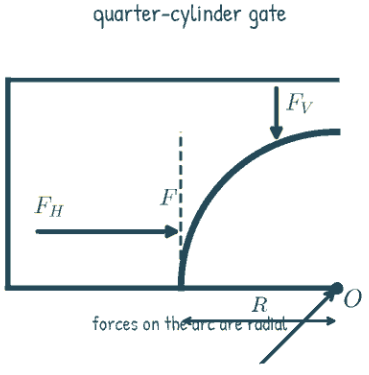
→ the vertical part needs the real or imagined column above the curved surface

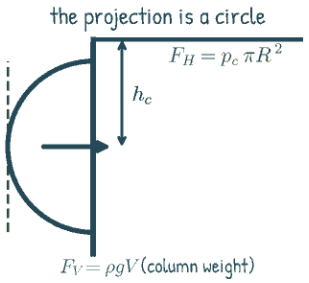
KEY RESULT
curved faces
F_H = p_c A_proj
F_V = ρgV of the column above
resultant through the centre of a circular arc

TRAP
wetting
Using the body volume instead of the liquid column.
Assuming F_V always points down: a face wetted from below feels upthrust.

CHECK
limits
Flat vertical face: F_V = 0, F_H = p_cA.
Horizontal face: F_H = 0, F_V = weight above.

Split a curved-face force: horizontal from the projection, vertical from the column weight





KEEP / split the force: projection for horizontal, column weight for vertical

What changes when the tank moves?

? Accelerate a cup and the surface tilts. The physics is still hydrostatics, with a different gravity.

1 **One substitution**

In the tank frame every element feels gravity plus the pseudo-force −ma. Define g_eff = g − a; the fluid is static under g_eff, so level surfaces are perpendicular to it and pressure grows along −g_eff.

$$\mathbf{g}_{eff} = \mathbf{g} - \mathbf{a}, \qquad \nabla p = \rho\, \mathbf{g}_{eff}$$

2 **The free surface tells you the answer**

A free surface is a level surface of p, so it is perpendicular to g_eff and therefore tilts by

$$\tan\theta = \frac{a}{g} \tag{3.4}$$

→ vertical acceleration adds: g_eff = g + a; free fall gives g_eff = 0 and no pressure gradient

3 **Volume, not the surface, fixes the level**

Before spilling, the water volume is unchanged. A plane cut through a rectangular tank keeps the volume only if it passes through the mid-length point, so the surface pivots there: rise at the back equals fall at the front.

$$\text{rise at back} = \frac{L}{2}\tan\theta$$

→ E5: brim at a = 1.63 m/s²; bottom exposed at 4.08 m/s², after spilling has begun

4 **Light and heavy bodies disagree**

Buoyancy points along −g_eff. A helium balloon, lighter than air, leans along a; a plumb line, denser than air, hangs along g_eff, against a. Both are the same vector statement applied to bodies of opposite density.

→ rule: balloon leans with a, plumb line against a

5 **A worked lift**

Q7: a tank accelerates upward at g. Effective gravity doubles, so the pressure at depth h doubles and so does the reading of a scale under the tank.

$$p = p_0 + 2\rho g h$$

→ in free fall g_eff = 0: a hole in a falling bottle does not squirt

KEY RESULT

moving vessels

$$\tan\theta = a/g$$

$$p = p_0 + \rho\, g_{eff} h$$

level surfaces perpendicular to g_eff

GATE

rigid body only

The whole vessel accelerates together; no sloshing, no rotation of parts.

For rotation add the centrifugal field ω²r radially.

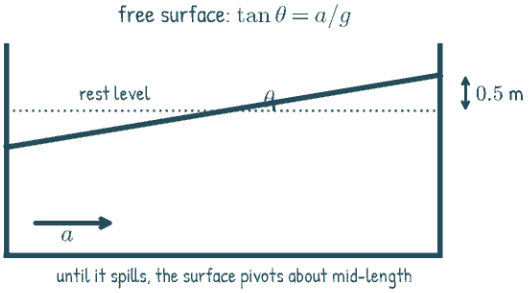
TRAP

volume bookkeeping

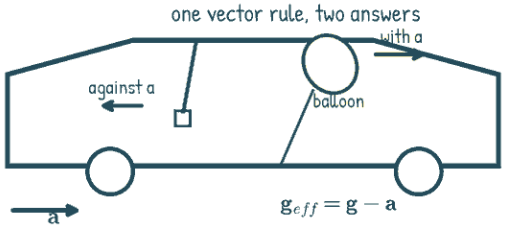
Using the pivot rule after water starts spilling.

Forgetting that a closed tank has no free surface, only tilted isobars.

Rigid acceleration is statics with g_eff: the balloon leans with a, a plumb line against it



until it spills, the surface pivots about mid-length



KEEP / replace g by g_eff = g − a and reuse every static tool

Why does the coffee climb the wall?

? Spin a bucket and the surface becomes a paraboloid. The shape follows from one force balance and one volume count.

1 Balance a surface element

A element on the surface feels N, mg and the pseudo-force $m\omega^2 r$. Dividing the horizontal balance by the vertical one gives the slope:

$$\tan\alpha = \frac{dz}{dr} = \frac{\omega^2 r}{g} \quad \Rightarrow \quad z = z_0 + \frac{\omega^2 r^2}{2g} \tag{3.4}$$

2 The pressure field behind it

In the rotating frame, $dp/dr = \rho\omega^2 r$ and $dp/dz = -\rho g$. Integrating gives a "rotating Bernoulli" whose constant-pressure surfaces are the same paraboloids.

$$p - \tfrac{1}{2}\rho\omega^2 r^2 + \rho g z = \text{const}$$

3 Volume conservation splits the difference

The paraboloid holds exactly half the volume of the cylinder that circumscribes it. So the liquid lost near the axis equals the liquid gained at the rim: the rest level sits halfway, and each side moves $\Delta z/2$.

$$\Delta z = \frac{\omega^2 R^2}{2g}, \qquad \text{centre drop} = \text{rim rise} = \frac{\Delta z}{2}$$

→ ω = 4 s^{-1}, R = 0.30 m: Δz = 7.35 cm total, 3.7 cm each way

4 When the bottom shows

The surface reaches the bottom centre when the centre drop equals the original depth. Beyond that the paraboloid is cut by the floor and the simple volume argument must be replaced by the actual paraboloid volume.

→ spilling begins when the rim rise exceeds the freeboard

5 How the numbers scale

The rim rise is quadratic in the radius. Halve the bucket and the same spin lifts the rim by a quarter as much.

$$\Delta z(0.15\ \text{m}) = \frac{16(0.0225)}{19.6} = 1.84\ \text{cm}$$

→ the centre drop is still half of that: volume conservation is untouched by the radius

KEY RESULT

rotating vessel

$$z = z_0 + \frac{\omega^2 r^2}{2g}$$

$$p - \tfrac{1}{2}\rho\omega^2 r^2 + \rho g z = \text{const}$$

rim-to-centre height = $\omega^2R^2/2g$

TRAP

two errors

Saying the whole surface rises by Δz: it pivots about the rest level.

Forgetting to conserve volume in a closed or partly filled vessel.

NUMBERS

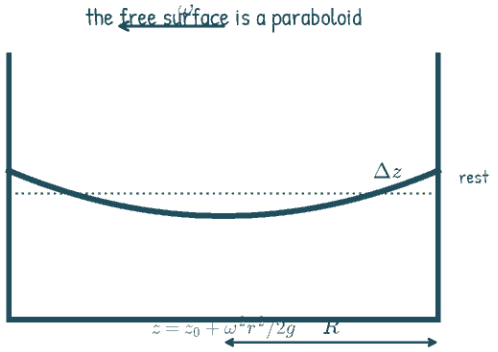
numbers

$\Delta z = \omega^2R^2/2g$

ω = 4, R = 0.30: 7.35 cm

each side 3.7 cm

Volume conservation splits the paraboloid: the centre falls exactly as much as the rim rises



$$z(r) = z_0 + \frac{\omega^2 r^2}{2g}$$

$$\text{rim rise} = \frac{\omega^2 R^2}{2g}$$

$$\text{centre drop} = \text{rim rise}$$

$$\omega = 4\ \text{s}^{-1},\ R = 0.30\ \text{m}$$

$$\Delta z = 7.4\ \text{cm, split } 3.7 + 3.7$$

KEEP / paraboloid: rim rise = centre drop when the volume is conserved

What does a floating body actually displace?

? Archimedes is a pressure integral in disguise, and the volume that matters is the one the fluid lost.

1 **Archimedes from the pressure integral**

The horizontal pressure forces on a submerged body cancel in pairs. The vertical ones leave the difference between the push on the bottom and on the top, which is exactly the weight of the displaced fluid.

$$F_b = (p_2 - p_1)A = \rho_f g V_{disp} \qquad (3.6)$$

2 **Floating is an equilibrium statement**

A floating body displaces just enough fluid to carry its weight, so the submerged fraction is the density ratio. In layered fluids each layer contributes its own ρgV.

$$\frac{V_s}{V} = \frac{\rho_b}{\rho_f}$$

→ wood at 667 kg/m³ floats with two thirds under water

3 **Apparent weight and the two-fluid case**

A body weighed while submerged seems lighter by the weight of the fluid it displaces. With two liquids the displaced volumes are set by the layer boundaries, not by the body shape alone.

$$W_{app} = W - \rho_f V g$$

→ a cylinder half in water, half in oil: ρb = ½(1000) + ½(800) = 900 kg/m³

4 **Where the buoyant force acts**

It acts at the centroid of the displaced volume, the centre of buoyancy B, not at the centre of mass. That distinction is the whole of the next page on stability.

→ a body whose density equals the fluid has no preferred depth: neutral equilibrium

5 **A worked two-fluid float**

Q8: a cylinder of density 667 floats in water, then oil of density 500 is poured until it is just covered. Two liquids now share the buoyancy.

$$667 = 1000f + 500(1 - f) \;\Rightarrow\; f = \tfrac{1}{3}$$

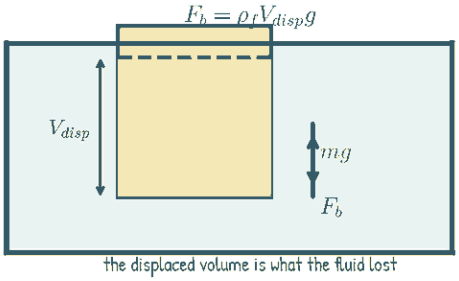
→ without the oil f = 2/3; the upper liquid carries the rest, so the cylinder rides higher

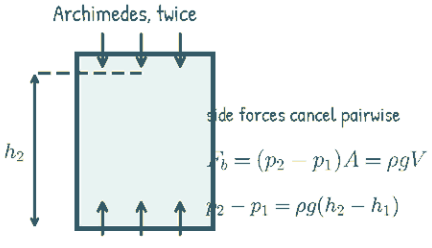
KEY RESULT
buoyancy
$F_b = \rho_f V_{disp} g$
$V_s/V = \rho_b/\rho_f$
B = centroid of the displaced volume

TRAP
volume confusion
Using the whole body volume for a floating body.
Putting F_b at the centre of mass, then wondering why the moments are wrong.
Ignoring air buoyancy in a precision weighing.

CHECK
limits
ρb -> ρf: the body is neutrally buoyant.
ρb > ρf: it sinks and F_b is fixed at ρfVg.
In an accelerating lift use g_eff.

Buoyancy is the pressure integral over the body, equal to the weight of displaced fluid





KEEP / buoyancy = weight of displaced fluid, acting at the centre of buoyancy

When does a floating body tip over?

? A ship is stable although its centre of mass is above the waterline. The metacentre explains why.

1 Two different points

Gravity acts at G, the centre of mass. Buoyancy acts at B, the centroid of the displaced volume. When the body heels, the submerged shape changes and B slides toward the low side; the body may right itself or capsize.

2 The metacentre

For a small heel the vertical through the new B meets the original centreline at M. The distance from B to M depends on the waterplane area and the displaced volume:

$$BM = \frac{I}{V}$$

→ I is the second moment of the waterplane area about the heel axis; V is the displaced volume

3 The stability criterion

If M is above G, a heel produces a righting moment; if M is below G, the moment tips the body further. For small angles the righting moment is linear in the heel.

$$GM = BM - BG > 0, \qquad \tau \approx W\,GM\,\theta$$

4 A cylinder, to see the scaling

A cylinder of radius r floating with draft d has I = πr⁴/4 and V = πr²d, so BM = r²/4d. A wide, shallow body is stable; a tall, narrow one is tender.

$$BM = \frac{r^2}{4d}$$

→ a fully submerged body has no waterplane: stability needs G below B

5 A worked metacentre

A cylinder of radius 0.50 m floats with a 0.50 m draft. Its metacentre sits above the centre of buoyancy by r²/4d.

$$BM = \frac{(0.50)^2}{4(0.50)} = 0.125\ \text{m}$$

→ if G is 0.10 m above B then GM = 0.025 m: stable, but tender

KEY RESULT

flotation stability

$$BM = I/V$$

$$GM = BM - BG$$

stable iff GM > 0

TRAP

two mistakes

Assuming a low G is enough: a submerged body needs G below B.

Using the volume of the whole body instead of the displaced volume in BM.

NUMBERS

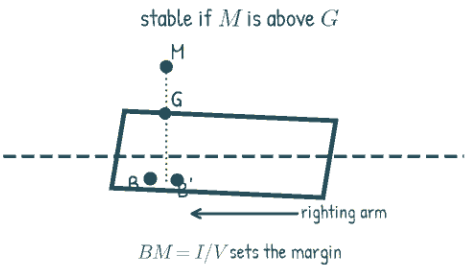
feel for sizes

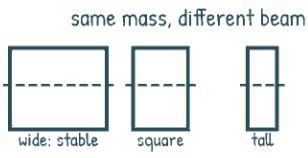
cylinder: BM = r²/4d

wide beam buys stability cheaply

ballast lowers G, not B

Stability is a metacentre question: BM = I/V, measured above the centre of buoyancy





KEEP / stable when the metacentre M is above G; BM = I/V

Does the level rise when the ice melts?

? Four versions of one question. One ledger answers all four without a single integral.

1 ## The ledger

A floating body displaces its own mass of liquid. After melting, the water it becomes occupies its own mass divided by the density of the liquid it joins. Compare those two volumes; the sign of the difference is the sign of the level change.

$$\Delta V = \frac{m}{\rho_f} - \frac{m}{\rho_{melt}}$$

2 ## Pure ice: exactly zero

Floating ice displaces m/ρw, and the melt water is m/ρw. The level does not move, however much of the cube is above water.

$$\Delta h = 0$$

3 ## Ice carrying a stone

While floating, the stone's weight is carried by the ice, so it displaces m_s/ρw. After melting the stone lies on the bottom and displaces only its own volume m_s/ρs. The difference is a fall.

$$\Delta V = m_s\left(\frac{1}{\rho_w} - \frac{1}{\rho_s}\right) > 0$$

→ E7: 0.1 kg stone in 0.02 m² cylinder -> 3 mm fall

4 ## Salt water and trapped air

In salt water the melt is fresh and lighter than the brine it displaced, so the level rises. An air bubble has no mass, so it changes nothing: the ice still displaces its own mass.

→ rules: stone falls, bubble unchanged, salt water rises, pure ice unchanged

5 ## A worked stone

A 10 cm ice cube carries a 20 g stone in a tank of area 0.02 m². The stone loses the extra displacement it had while riding the ice.

$$\Delta h = \frac{m_s(1/\rho_w - 1/\rho_s)}{A} = \frac{(0.02)(6\times10^{-4})}{0.02} = 0.6\ \text{mm}$$

→ without the stone the level does not move at all

KEY RESULT

ice family

pure ice: unchanged

ice + stone: falls

ice + air: unchanged

ice in salt water: rises

TRAP

the usual slip

Tracking volumes of ice rather than masses of water.

Forgetting that a sunken stone still displaces its own volume.

CHECK

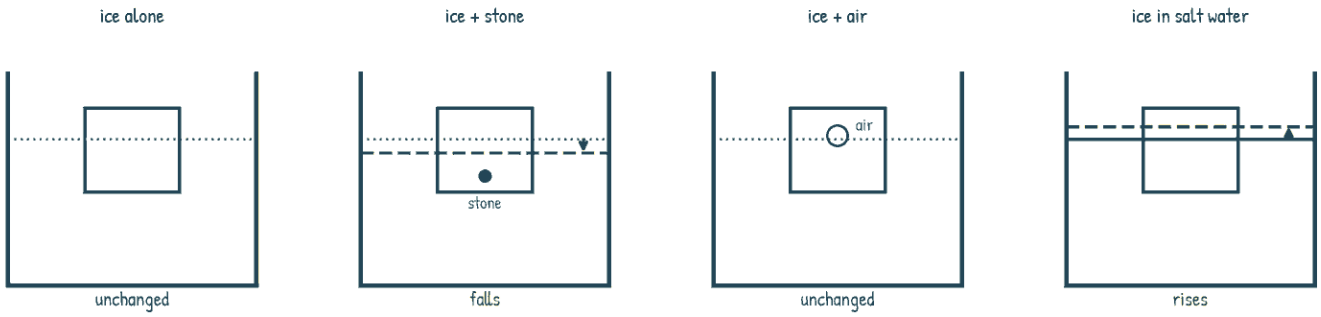
one-line audit

Floating: displaced volume ∝ mass.

Sunk: displaced volume ∝ volume.

Melt: added volume = mass / ρ of the receiving liquid.

The displaced-volume ledger answers every melting-ice variant in one line



KEEP / compare the floating displacement with the volume added after melting

When may you use Bernoulli?

? The equation is short and the conditions are strict. Most lost marks are gate violations, not algebra.

1 **Derive it once, on a streamtube**

Take a steady streamtube between sections 1 and 2. In a time dt the pressure at section 1 does work $p_1 A_1 v_1 dt$ and section 2 removes $p_2 A_2 v_2 dt$. The difference changes the kinetic and potential energy of the fluid between the sections.

$$p_1 + \tfrac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \tfrac{1}{2}\rho v_2^2 + \rho g y_2 \qquad (3.8)$$

2 **Divide by ρg and it becomes heads**

Each term is now a length: pressure head, velocity head, elevation head. A pitot tube reads the first two; a Venturi turns one into the other.

$$\frac{p}{\rho g} + \frac{v^2}{2g} + y = \text{const}$$

3 **The four gates**

Steady flow, incompressible fluid, no viscous work, and both points on the same streamline. Drop any one and the constant changes along the flow, or between streamlines.

→ a long pipe fails gate 3; a pump changes the constant; a boundary layer fails gate 4

4 **Where it must not be used**

Across a long narrow pipe the pressure falls linearly with length, not with v². Across a pump or turbine there is shaft work. In a viscous wake the total head drops from point to point.

→ if the question names viscosity, length or flow rate, suspect Poiseuille

5 **A worked pipe**

Q12: water at 1.2 m/s in a 2 cm radius pipe enters a 1 cm radius section 10 m lower. Continuity fixes the speed; Bernoulli fixes the pressure.

$$v_2 = 1.2\left(\tfrac{2}{1}\right)^2 = 4.8 \text{ m/s}$$

$$p_1 - p_2 = \tfrac{1}{2}\rho(v_2^2 - v_1^2) + \rho g(10) = 1.088 \times 10^5 \text{ Pa}$$

→ the 10 m drop contributes 98 kPa; the narrowing only 10.8 kPa

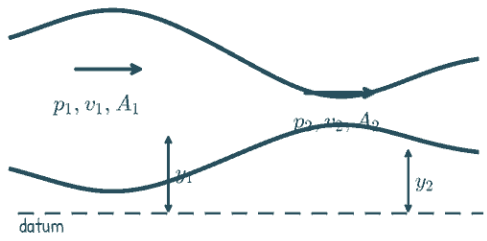
GATE
the four gates
1 steady
2 incompressible
3 inviscid
4 along one streamline

KEY RESULT
heads
$\frac{p}{\rho g} + \frac{v^2}{2g} + y = \text{const}$
a pump adds head; a turbine removes it

TRAP
the classic
Applying Bernoulli between the ends of a long pipe and calling the loss a pressure drop.
Using it across streamlines in a vortex.

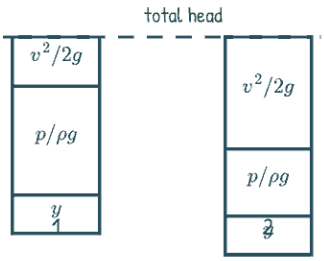
Bernoulli is work-energy for a streamtube, drawn as stacked heads

work-energy on a streamtube



$$p + \tfrac{1}{2}\rho v^2 + \rho g y = \text{const}$$

the split changes, the total does not



KEEP / name the four gates before you write the equation

How do Venturi, pitot and siphon read a pressure difference?

? Three instruments, one balance. Each converts a speed into a height you can see.

1 ## Venturi: continuity plus Bernoulli

The throat is faster by the area ratio. Bernoulli between the wide section and the throat gives the pressure drop; a manometer reads it as a height difference.

$$Q = A_1\sqrt{\frac{2\Delta p}{\rho(A_1^2/A_2^2 - 1)}} \tag{3.9}$$

→ A1/A2 = 2 gives v2 = 4v1 and a dynamic-pressure rise of 15/2 ρv1²

2 ## Pitot: bring the fluid to rest

At the mouth of a pitot tube the fluid stagnates, so the pressure there is the static pressure plus ½ρv². The difference is measured directly.

$$v = \sqrt{\frac{2\Delta p}{\rho}}$$

→ E8: Δp = 1470 Pa, A1/A2 = 2 -> Q = 1.77×10⁻³ m³/s

3 ## Siphon: gravity across two free surfaces

Both ends are at atmospheric pressure, so the speed depends only on the height difference between the surfaces. The top of the tube is the constraint: its pressure must stay above the liquid's vapour pressure.

$$v = \sqrt{2g\,\Delta h}\,, \qquad p_{top} = p_0 - \rho g h_{top} - \tfrac{1}{2}\rho v^2 \geq 0$$

→ this caps the lift at about 10.3 m of water

4 ## Atomiser and aspirator

A fast stream has low pressure, so it draws liquid up a side tube and shatters it into droplets. Same balance, run backwards: high speed, low pressure, entrainment.

5 ## A worked siphon

Q14: a siphon lifts water 3 m above the supply surface and discharges 2 m below it. The driving head is the 2 m drop, not the 3 m lift.

$$v = \sqrt{2(9.8)(2)} = 6.26\ \text{m/s}$$

$$p_{top} = p_0 - \rho g(3) - \tfrac{1}{2}\rho v^2 = 5.2 \times 10^4\ \text{Pa}$$

→ positive, so this siphon works; at a 10 m lift it would not

KEY RESULT

three readings

Venturi: Q from Δp and the area ratio

Pitot: v from the stagnation pressure

Siphon: v from the free-surface height difference

TRAP

manometer fluid

Using Δp = ρgΔh with the wrong density: a mercury manometer reads (ρHg − ρ)gΔh.

Ignoring the vapour-pressure cap on a siphon.

CHECK

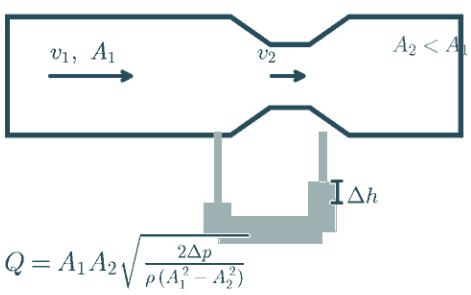
limits

A2 -> A1: Q -> ∞ is the signal that the model has lost the pressure drop.

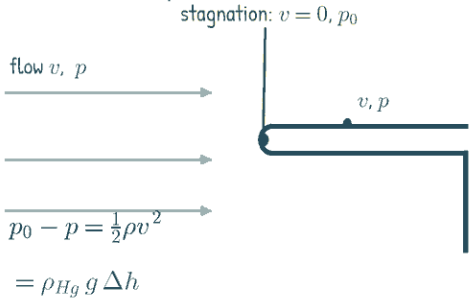
Pitot in a viscous boundary layer reads too little.

Venturi and pitot are one Bernoulli balance read off a manometer

Venturi: the throat reads the flow



Pitot: stop it, read the head



KEEP / continuity supplies the speed ratio; Bernoulli supplies the pressure drop

How fast does water leave a hole, and where does it land?

? Depth sets the speed, the fall sets the time, and their product sets the range.

1 ## Speed from the surface

Apply Bernoulli between the free surface (p_0, $v \approx 0$) and the hole (p_0). The depth h is the only head left. The exact result keeps the surface speed through continuity.

$$v = \sqrt{2gh} \quad \left(\text{exact}: \; v = \sqrt{\frac{2gh}{1-(a/A)^2}}\right) \tag{3.10}$$

→ the correction is 0.5% for a/A = 0.1, but 25% for a/A = 0.6

2 ## Range: two motions at once

The jet leaves horizontally and falls a distance H − h. Eliminating time gives the range, whose maximum is at half depth.

$$x = 2\sqrt{h(H-h)}, \qquad x_{max} = H \text{ at } h = H/2 \tag{3.11}$$

3 ## Draining: a clock made of area

With a ≪ A the flow is quasi-steady: $-A\,dh/dt = a\sqrt{2gh}$. Separating and integrating gives a finite emptying time, and a height that is a perfect square in time.

$$T = \frac{A}{a}\sqrt{\frac{2H}{g}}, \qquad h(t) = H\left(1 - \frac{t}{T}\right)^2$$

→ A/a = 100, H = 1.25 m -> T = 50.5 s

4 ## Two holes, moving tanks and siphons

Two holes at depths h_1 and h_2 give speeds in the ratio $\sqrt{h_1}:\sqrt{h_2}$ and thrusts in that ratio squared. A moving tank adds a pseudo-force term; a sealed tank slows as the air above expands.

→ a jet from a hole in a tank on a cart pushes the cart the other way

5 ## A worked range

Q13: a tank 1.25 m deep has a hole 0.8 m below the surface. The jet falls the remaining 0.45 m while travelling horizontally.

$$x = 2\sqrt{(0.8)(0.45)} = 1.2 \text{ m}$$

→ Q15: A/a = 100 gives T = 50.5 s, and the tank is exactly empty at T

KEY RESULT

Torricelli family

$$v = \sqrt{2gh}$$

$$x = 2\sqrt{h(H-h)}$$

$$T = \frac{A}{a}\sqrt{\frac{2H}{g}}$$

GATE

quasi-steady

a ≪ A, so the surface speed is negligible.

The hole is sharp and short; a long tube makes it a Poiseuille problem.

TRAP

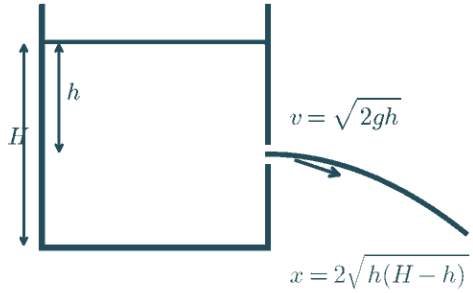
surface speed

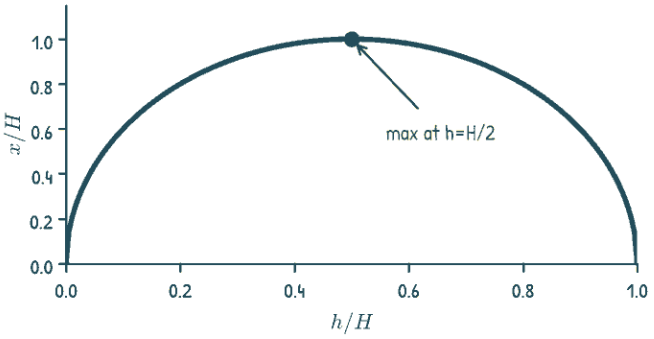
Forgetting the 1 − (a/A)² correction for a large hole.

Using H instead of H − h in the range formula.

Torricelli gives the speed; the parabola gives the range

Torricelli: depth becomes speed





KEEP / v = sqrt(2gh); range peaks at half depth; the tank empties in finite time

Why does a hose push back?

? Pressure is not the force on a plate. Momentum flux is.

1 **Mass flow carries momentum**

A jet of area a and speed v delivers mass ρav every second, each kilogram carrying v. If the plate stops the jet, the whole momentum flux becomes a force.

$$F = \dot{m}v = \rho a v^2 \tag{3.12}$$

→ a 5 cm² hose at 20 m/s pushes with 200 N

2 **A plate that runs away**

If the plate moves at u in the jet direction, the impact speed is v − u and the mass striking it per second is ρa(v − u). The force falls as the square of the relative speed, and the power delivered peaks at u = v/3.

$$F = \rho a (v-u)^2, \qquad P = Fu$$

3 **Curved vanes and reversals**

A vane that turns the jet through 180° reverses its momentum, so the force doubles. Turning it through an angle φ gives a factor (1 − cosφ) per unit mass flow.

$$F = \rho a v^2 (1 - \cos\varphi)$$

4 **Ejection is the same statement**

A rocket or a sprinkler feels the reaction of the momentum it throws away. The thrust is the momentum flux plus any pressure difference across the nozzle exit.

$$F = \rho a v_e^2 + (p_e - p_0) A_e$$

→ for a hose the pressure term is already counted inside ρav² if the jet is at atmospheric pressure

5 **A worked hose**

Q16: a 5 cm² hose delivers water at 20 m/s onto a wall, and the water leaves along the wall. All the horizontal momentum flux becomes force.

$$F = \rho a v^2 = 1000(5 \times 10^{-4})(400) = 200 \text{ N}$$

→ Q17: holes at h and 4h give speeds 1:2 and thrusts 1:4

KEY RESULT

momentum flux

$$F = \rho a v^2$$

$$F = \rho a (v-u)^2$$

reversal doubles the force

TRAP

not pressure

Using pA at the nozzle instead of the momentum flux.

Forgetting the relative speed for a moving plate.

Adding speeds instead of subtracting them.

CHECK

limits

u = 0 recovers the fixed plate.

u = v gives zero force and zero power.

φ = 180° doubles the force.

A jet pays momentum flux rho a v squared; a moving plate pays less



KEEP / force = momentum flux: ρav² stopped, ρa(v−u)² on a retreating plate

Why does a narrow pipe strangle the flow?

? Viscosity turns a pressure difference into a flow rate, and the fourth power of the radius decides who wins.

1 ## Newton's law of viscosity

Fluid layers slide past each other; the shear stress is proportional to the velocity gradient. The constant η is the dynamic viscosity, in Pa·s. At a solid wall the fluid does not slip.

$$\tau = \eta \frac{dv}{dy}$$

2 ## Shell balance, then integrate

On a cylindrical shell of radius r and length L, the pressure force $\pi r^2 \Delta p$ balances the shear on the curved surface $2\pi r L \tau$. Substituting Newton's law and integrating with v(R) = 0 gives the parabolic profile.

$$\tau = \frac{r\Delta p}{2L}, \qquad v(r) = \frac{\Delta p}{4\eta L}(R^2 - r^2)$$

$$Q = \frac{\pi R^4 \Delta p}{8\eta L} \tag{3.13}$$

3 ## What the quartic does

Halve the radius and the same pressure drives 1/16 of the flow. The mean speed is half the centreline speed, and the wall shear is $4\eta v_{avg}/R$. Doubling the length doubles the drop.

$$v_{max} = 2v_{avg}, \qquad \tau_w = \frac{4\eta v_{avg}}{R}$$

4 ## Where the law stops

The derivation assumes laminar, steady, fully developed flow of a Newtonian fluid with no slip. Above Re ≈ 2000 the flow becomes turbulent, the profile flattens and Δp grows roughly as Q^2.

→ entrance length ≈ 0.06 Re D: near the inlet the profile is still developing

5 ## A worked radius, and a warning

A sphere of density 2000 falls at 0.1 m/s in glycerine (1260 kg/m³, 1.5 Pa·s). Solve Stokes for the radius, then test the answer.

$$r = \sqrt{\frac{9\eta v}{2\Delta\rho g}} = 9.65 \times 10^{-3}\ \text{m}, \qquad \text{Re} = 1.6$$

→ Re > 1, so Stokes is not valid here: the printed 3 mm answer in the source fails its own check

KEY RESULT

pipe flow

$$Q = \frac{\pi R^4 \Delta p}{8\eta L}$$

$$v_{max} = 2v_{avg}$$

wall shear 4ηv/R

GATE

laminar only

Re = ρvD/η ≲ 2000

Newtonian, no slip, fully developed

straight pipe, no fittings

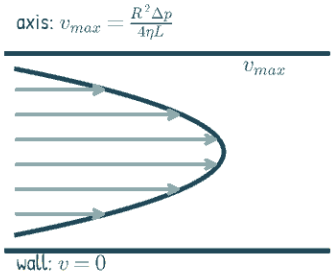
TRAP

two habits

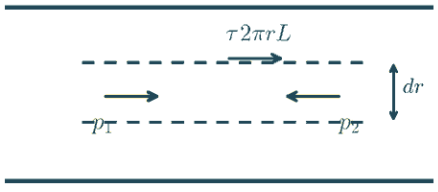
Using Bernoulli along a long pipe.

Forgetting that a narrower section has a much larger resistance, not just a larger speed.

Shell balance gives the parabola, then Poiseuille R to the fourth



the profile is a paraboloid, not a plug



$$\pi r^2 (p_1 - p_2) = 2\pi r L\, \tau$$

then integrate with no slip

KEEP / Q ∝ $R^4\Delta p/\eta L$: the radius is the whole design

How do pipes combine like resistors?

? Once the flow is linear in Δp, a pipe network is a circuit with pressure for voltage.

1 ## The hydraulic Ohm's law

Poiseuille is linear in the pressure drop, so each pipe is a resistor. The analogy is exact while the flow stays laminar and the fluid incompressible.

$$\Delta p = R_{hyd} Q, \qquad R_{hyd} = \frac{8\eta L}{\pi R^4}$$

2 ## Series and parallel

In series the same Q passes through every pipe and the drops add. In parallel every branch sees the same drop and the flows add, so conductances add.

$$R_{series} = \sum R_i, \qquad \frac{1}{R_{parallel}} = \sum \frac{1}{R_i}$$

3 ## A factor of sixteen, then seventeen

Two capillaries of radii r and r/2 in series carry the same flow. The narrow one has 16 times the resistance, so it takes 16/17 of the total drop: 2.82×10⁵ Pa out of 3×10⁵ Pa.

$$\frac{\Delta p_{narrow}}{\Delta p_{total}} = \frac{16}{17}$$

4 ## Design consequences

A small narrowing dominates a network; a parallel bypass steals flow in proportion to R^{-1}. The same algebra sizes blood vessels, irrigation lines and oil pipelines, until turbulence or elasticity breaks the linearity.

→ two equal pipes in parallel carry half each; a 2:1 radius pair carries 16:1

5 ## A worked divider

Q19: two capillaries of radii r and r/2, same length, in series under 3×10⁵ Pa. The resistances are in the ratio 1:16.

$$\Delta p_{narrow} = \tfrac{16}{17}(3 \times 10^5) = 2.82 \times 10^5 \text{ Pa}$$

→ the flow is the same in both; only the pressure drops differ

KEY RESULT

network rules

$$R = 8\eta L / \pi R^4$$

series: R adds, Q same

parallel: 1/R adds, Δp same

TRAP

analogy limits

Turbulent pipes: $\Delta p \propto Q^2$, so R is not constant.

Fittings, bends and valves add local losses.

Compliant vessels store volume, like capacitors.

NUMBERS

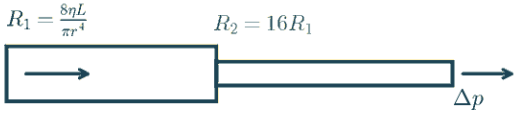
the 17

r and r/2: resistance ratio 16

narrow drop = 16/17 of total

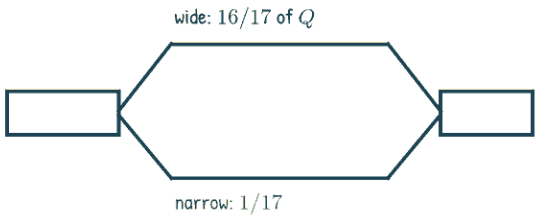
its flow if parallel = 1/17

Hydraulic Ohm: series add, parallel split as the fourth power of radius



series: resistances add

$$R = R_1 + R_2$$



parallel: same head, R splits as r to the fourth

$$Q_{narrow}/Q = \frac{1}{1+2^4}$$

KEEP / series adds R; parallel splits Q as 1/R

How fast does a small sphere sink?

? Stokes drag is a low-Reynolds result. The number that validates it is the one most answers omit.

1 **Three forces at terminal speed**

Weight down, buoyancy up, drag up. At terminal speed the acceleration is zero. Stokes solved the slow-flow drag around a sphere:

$$F_d = 6\pi\eta r v \tag{3.14}$$

→ the derivation assumes Re ≪ 1, so inertia is negligible and the flow is reversible

2 **Solve for the terminal speed**

Setting 6πηrv equal to the net weight 4/3πr³(ρ − σ)g gives a speed proportional to r². That squared signature is the experimental fingerprint of Stokes flow.

$$v_t = \frac{2r^2(\rho-\sigma)g}{9\eta} \tag{3.15}$$

→ a 0.10 mm raindrop: v_t = 1.21 m/s on paper

3 **Then check the Reynolds number**

The same raindrop has Re ≈ 16, far outside Stokes. The honest terminal speed is lower once quadratic drag enters, of order 0.7 to 1 m/s. An answer without Re is an unfinished answer.

$$\mathrm{Re} = \frac{\rho v D}{\eta}, \qquad \text{Stokes needs } \mathrm{Re} \lesssim 1$$

4 **What replaces it at larger sizes**

At large Re the drag is quadratic, F ≈ ½C_dρπr²v², and the terminal speed scales as √r instead of r². Between the two regimes there is a crossover radius; the Olympiad page finds it.

$$F_d \approx \tfrac{1}{2} C_d \rho \pi r^2 v^2$$

5 **The raindrop, checked**

E9: a 0.10 mm water drop in air. Stokes gives a speed, but the same numbers give a Reynolds number far above one.

$$v_t = 1.21\ \mathrm{m/s}, \qquad \mathrm{Re} = 16$$

→ the honest terminal speed is 0.7–1 m/s: Stokes overestimates once inertia matters

KEY RESULT

Stokes

$$F_d = 6\pi\eta r v$$

$$v_t = \frac{2r^2\Delta\rho g}{9\eta}$$

always print Re with v_t

GATE

Re ≲ 1

small sphere, slow flow, no wall nearby

wall correction: divide by 1 + 2.4r/R

above Re ≈ 1 use quadratic drag

TRAP

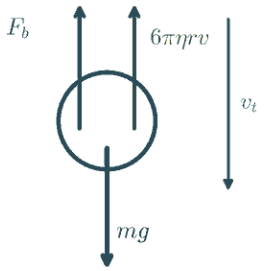
the missing check

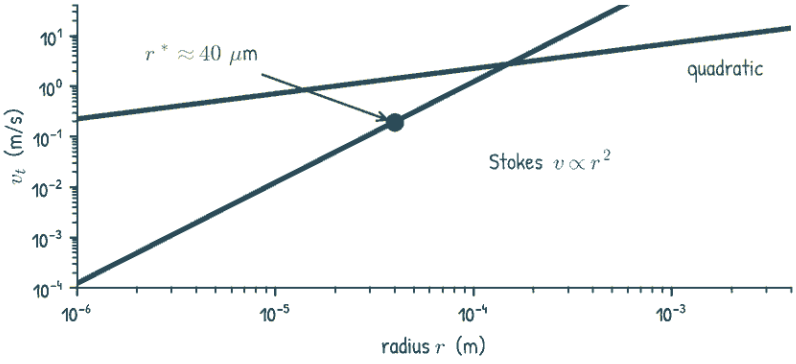
Quoting 1.21 m/s for a raindrop and stopping there.

Using the sphere radius where the diameter belongs in Re.

Stokes holds below Re about 1; a rain drop is already quadratic

steady fall: three forces





KEEP / terminal velocity v_t = 2r²Δρg/9η, valid while Re ≲ 1

Why is a soap film a stretched sheet?

? Surface tension can be read as a force per length or an energy per area. The film makes the factor of two visible.

1 ## Two definitions

Pull a slider along a soap film: the force per unit length of the slider is Y. Alternatively, count the energy needed to create new surface: Y is also the energy per unit area. For a simple liquid the two numbers are identical.

$$\gamma = \frac{F}{L} = \frac{W}{\Delta A}$$

→ units: N/m = J/m²

2 ## The film doubles everything

A soap film has two free surfaces, so the force on the slider is 2YL and stretching it by dx costs 2YLdx. This factor of two is the most reliably forgotten factor in the chapter.

$$F_{film} = 2\gamma L, \qquad W = 2\gamma \Delta A$$

3 ## Where the energy goes

A molecule in the bulk is pulled equally in all directions; one at the surface has fewer neighbours and therefore higher energy. Creating area raises the energy, which is why free surfaces shrink to the smallest area they can.

$$U_{surface} = \gamma A$$

→ splitting a 1 mm drop into 1000 droplets costs 8.2×10^{-4} J

4 ## Values and temperature

Water's Y falls from 0.0728 N/m at 20 °C toward zero at the critical point; mercury's is about 0.465 N/m, which is why it beads and does not wet glass. Surfactants lower Y by sitting at the interface.

5 ## Two worked surfaces

Q24: lift a 10 cm square frame from water, film on both faces. Q25: split a 1 mm drop into a thousand equal droplets.

$$F = 2\gamma(0.40) = 0.058 \text{ N}, \qquad W = \gamma(36\pi R^2) = 8.2 \times 10^{-6} \text{ J}$$

→ both use the two-surface count; forgetting it halves or doubles the answer

KEY RESULT

surface tension

$$\gamma = F/L = W/\Delta A$$

$$F_{film} = 2\gamma L$$

U = YA for one surface

TRAP

factor two

Using YL for a soap film.

Counting the area of one face when two are created.

Mixing up N/m and N/m².

NUMBERS

values at 20 °C

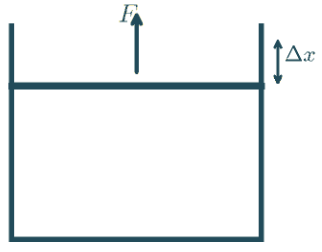
water 0.0728 N/m

mercury 0.465 N/m

soap solution ≈ 0.025–0.03 N/m

Surface energy is the same number as surface tension, counted per face

a film has two faces



$F = 2\gamma L$

pull the wire by dx: work = 2 gamma L dx

energy and force agree

$$W = 2\gamma L\, \Delta x = \gamma\, \Delta A$$



two surfaces, each pulls with gamma L

KEEP / two definitions, one number; a film has two faces

Why is the pressure inside a bubble higher?

? Curvature costs surface energy, so a curved interface must carry a pressure step.

1 ## Balance half a drop

Cut a droplet through its centre. The pressure difference pushes the hemisphere out with $\Delta p \cdot \pi r^2$; the rim pulls it back with $2\pi r Y$. Dividing gives the excess pressure.

$$\Delta p = \frac{2\gamma}{r}$$

2 ## Count surfaces before you divide

A soap bubble has an inner and an outer surface, so the same balance carries a factor two. A cylindrical jet has one curved radius, and the missing second curvature halves the result again.

$$\Delta p_{bubble} = \frac{4\gamma}{r}, \qquad \Delta p_{jet} = \frac{\gamma}{r} \tag{3.17}$$

3 ## The general law

For any smooth patch with principal radii r_1 and r_2, the pressure step is Y times the sum of the curvatures. A saddle has curvatures of opposite sign and can have zero mean curvature.

$$\Delta p = \gamma\left(\frac{1}{r_1} + \frac{1}{r_2}\right) \tag{3.18}$$

4 ## Consequences

Small bubbles have the higher pressure, so they empty into large ones through a connecting tube. Blowing a bubble is hardest at the start, when r is smallest. Two equal bubbles coalesce slowly; a small one disappears.

→ Q21: bubbles r and 3r coalesce isothermally to R = 2r, conserving moles, not volume

5 ## Bubbles that merge

Q21: two soap bubbles of radii r and 3r coalesce isothermally in vacuum. For a bubble $pV \propto r^2$, so the moles add as radii squared.

$$r^2 + (3r)^2 = R^2 \;\Rightarrow\; R = \sqrt{10}\, r$$

→ the chapter prints R = 2r from an arithmetic slip in the second bubble: see SOURCE-AUDIT

KEY RESULT

excess pressure

$$\Delta p = 2\gamma/r$$

$$\Delta p = 4\gamma/r$$

$$\Delta p = \gamma(1/r_1 + 1/r_2)$$

TRAP

surface counting

Using 2Y/r for a soap bubble.

Using 4Y/r for a liquid drop in air.

Forgetting that a bubble in water has one surface; in air it has two.

CHECK

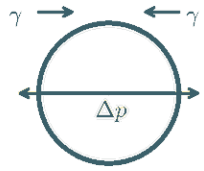
limits

$$r \to \infty : \; \Delta p \to 0$$

sphere: both radii equal

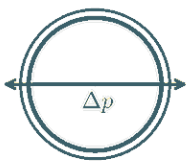
cylinder: one radius infinite

Excess pressure counts surfaces and principal curvatures



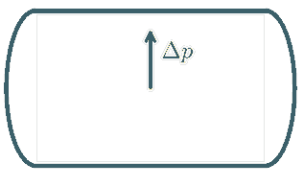
drop: one surface

$\Delta p = 2\gamma/r$



bubble: two surfaces

$\Delta p = 4\gamma/r$



jet: one curvature

$\Delta p = \gamma/r$

KEEP / drop 2Y/r, soap bubble 4Y/r, cylinder Y/r

How high does water climb a narrow tube?

? Capillary rise is a force balance and an energy minimum, and the two disagree by a factor of two unless you audit the dissipation.

1 ## Force balance on the column

The rim pulls up with $2\pi r\gamma\cos\theta$; the column weighs $\rho g\pi r^2 h$. Setting them equal gives Jurin's law. The contact angle is measured through the liquid.

$$h = \frac{2\gamma\cos\theta}{\rho g r} \tag{3.19}$$

→ r = 0.5 mm, clean water: h = 2.97 cm

2 ## Mercury walks the other way

For $\theta > 90^\circ$ the cosine is negative and the surface is depressed: a convex meniscus. Mercury in a 1 mm tube sits 1.07 cm below the outside level.

$$h = \frac{2(0.465)\cos 140^\circ}{(13600)(9.8)(5\times10^{-4})} = -1.07\ \text{cm}$$

3 ## Energy: the same answer, twice the budget

The potential energy is $\tfrac{1}{2}\rho g\pi r^2 h^2$ minus the surface energy released $2\pi r\gamma h\cos\theta$. Minimising reproduces Jurin exactly. But the total adhesion release is twice the gravitational gain; the other half is dissipated as the column oscillates and settles.

$$U(h) = \tfrac{1}{2}\rho g\pi r^2 h^2 - 2\pi r\gamma h\cos\theta$$

→ E10: $U_g = 3.4\times10^{-4}$ J, adhesion release 6.8×10^{-4} J

4 ## Short and tilted tubes

If the tube is shorter than h, the meniscus flattens until its radius R' satisfies $hR = h'R'$; the liquid does not overflow. Tilting the tube lengthens the wetted run but leaves the vertical rise unchanged.

$$hR = h'R'$$

5 ## A worked short tube

Q22: water rises 12 cm in a capillary, but the tube is cut to 9 cm. The meniscus flattens until its radius of curvature satisfies the same force balance.

$$hR = h'R' \;\Rightarrow\; \frac{R'}{R} = \frac{12}{9} = \frac{4}{3}$$

→ Q23: mercury in a 1 mm tube is depressed 1.07 cm, because cos140° is negative

KEY RESULT

capillarity

$$h = \frac{2\gamma\cos\theta}{\rho g r}$$

$$hR = h'R'$$

θ measured through the liquid

TRAP

signs and factors

Measuring θ through the gas, which flips the sign for mercury.

Using the diameter where the radius belongs.

Assuming a short tube overflows.

NUMBERS

bench numbers

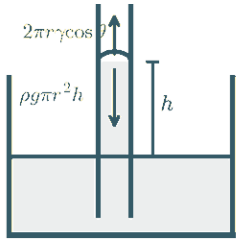
water: 2.97 cm at r = 0.5 mm

mercury: −1.07 cm at r = 0.5 mm

$h \propto 1/r$, so 10 μm gives 2.97 m

Jurin balances rim force against weight; a short tube only changes R

a narrow tube lifts a tall column



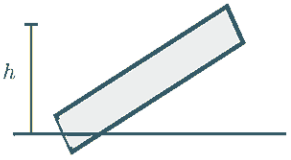
rim pull balances the column weight

short tube: R grows, h falls



$hR = h'R'$

tilt: same vertical h



KEEP / h = 2γcosθ/ρgr; a short tube only changes the curvature